

# Visualization and Manipulation of Air Conditioner Flow via Touch Screen

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## ABSTRACT

Though air conditioners (A/Cs) have long been a part of homes and businesses alike, the interfaces for adjusting and managing these devices have not changed significantly over the years. In particular, directional A/Cs such as wall mounted, window, or split units commonly found in smaller homes or Asian countries still predominantly require button-based remotes for adjusting the direction and positioning of air vents and flow direction.

In this paper, we present a smartphone controlled interface for both manipulating A/C air flow and visualizing the resulting output as a 3D Augmented Reality (AR) overlay. In contrast with previous work, we generate airflow models based on the A/C's exhaust vents, allowing users to see the effects of interactions on airflow in real time. We also implemented and tested three different methods for manipulation including, swipe, drag and button based manipulations using finger or device gestures. Experiments showed that participants (N=50) were able to control air flow most quickly with the swipe based interface, which outperformed the drag and button median completion times by 26.5% and 36.1%, respectively.

**Keywords:** Visualization, augmented reality, 3D user interfaces, interaction, comfort, air conditioning

## 1 INTRODUCTION AND RELATED WORK

Many air conditioners (A/Cs), especially those designed for larger rooms or spaces, are designed to generate uniform air conditions throughout the entire room. However, personal thermal comfort varies greatly from individual to individual, and if there are multiple people in a room, their comfort is likely to differ greatly. Our goal in this work is to enable better control of air conditioning systems such that individual preferences can be achieved. We want to accomplish this by 1) enabling quick, accurate control of air flow and 2) ensure that users can view their changes without having to wait for the air conditioning system to adjust. Meeting these goals will improve individual thermal comfort and help control air flow in rooms of various sizes and numbers of inhabitants.

Air flow visualization has often been utilized for designing the thermal environment of a building. For example, Fukuda et al. [2] proposed an AR based system for visualizing thermal characteristics. Zhu et al. [6] also developed an airflow visualization system for building renovation. In this system, airflow is represented with stream lines and the color of lines represent temperature. To visually confirm the result of renovated airflow, this method utilized a CFD simulation with small sized room. This was selected to reduce computational costs, but it still took tens of minutes for the CFD simulation to complete. Lin et al. [4] also proposed a thermal environment visualization system based on AR with a mobile device. Manipulation techniques exist that make use of touch, gestures, and smartphones such as those in et al. [1, 5], but these have not been developed for or tested with airflow manipulation.

We have designed an interface that combines real-time visualization of an air conditioner's output trajectory with smartphone-based control of resulting air flow trajectory output vent direction in a

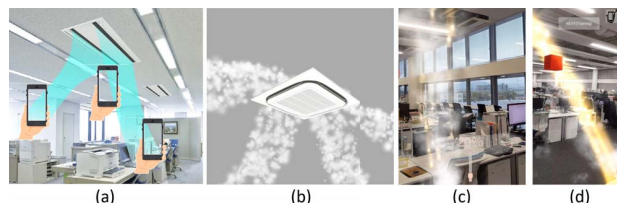


Figure 1: Images showing the air flow from directional air conditioner units as visualized from our system, including (a) a diagram outlining the AR visualization, (b) a bottom-up view of the simulated air flow in Unity coming from a 4-direction room unit, and screenshots from a smartphone screen showing a side view from the camera perspective during use in (c) an office and (d) with visualization.

single interface, as shown in Figure 1. In addition, we wanted to test what types of interaction would be the most effective to control air flow using a smartphone. For this purpose, we developed three different methods for controlling air flow, including *swipe*-, *drag*-, and *button*-based manipulation. Each of these methods was implemented within the smartphone's AR video see-through mode, and was tested in office environments of different sizes.

By controlling air flow volume, direction, and temperature, users should more effectively be able to improve the average thermal comfort of people in the room. To determine how well users could control air flow with our system and the three methods for control, we conducted an experiment with 50 participants (18 female, 32 male) to determine how well users were able to align the air flow with target points in the room. Results showed that the swipe gesture outperformed the other methods in terms of completion time, but that only slight (insignificant) differences were present in subjective evaluations.

## 2 SMARTPHONE AIR FLOW MANIPULATION

The primary components of our A/C control interface include the interactive visualization and touch-based user interface. Both of these were implemented in a single smartphone application, which provided a familiar, easy-to-use device for the user.

### 2.1 Visualization

The first step in our application was to design a visualization that would allow users to see the trajectory of air flow into the room. We implemented tracking using Vuforia v8.0.10, a platform for implementing video-based AR applications and used manual selection of the ceiling plane to initialize tracking. We also built replicas of the air conditioning units present in nearby office rooms in Unity v2018.3.11f1, as shown in (b) of Figure 1.

In order to visualize flow on a smartphone, we used a particle effect for rendering in combination with Unity's physics system, as shown in (b), (c), and (d) of Figure 1. Taking the frictional force of air into account, we set the particles to descend at a rate of 0.3 to 0.5 times gravity. This particle effect was created with a texture sheet animation, with a particle rate of 300 particles per second and a bounce of 0.25. These parameters were logically selected through testing and refinement with a small group of potential end-users.

Finally, finger presses are augmented with a "guide flow" visualization to show the resulting air flow trajectory based on simulated air flow, as shown in (d) of Figure 1.

## 2.2 Swipe, Drag, and Button Design

For airflow manipulations, we implemented three different methods to control the standard horizontal and vertical vent angle increments that can be set using an A/C remote control.

To match the real angular range of A/C guide fins, we set the angular adjustment intervals to 12 degrees vertical (60 degree range) and 18 degrees horizontal (45 degrees either left or right of the forward transform of the fins). These intervals were accurate enough to intersect the targets at any possible position in the room.

The first interface we implemented was *swipe*. Swipe gestures have long been used for control of various interfaces, including map scrolling, volume control, and selection of subsequent windows or pages [3]. As such, we thought this would be a logical choice for airflow control, as users could directly select a target location for the desired airflow output. To engage this method, the user places his or her finger on the screen and then moves it in the desired new airflow direction. Removing the finger from the screen subsequently halts the change in movement of the air flow.

The *drag* method allows the user to move the phone itself to control air flow. He or she first places a finger on the screen to activate the movement. Then, while holding a finger on the screen, the user moves the device itself, subsequently moving the selected finger position and air flow position at the same time. This touch essentially functions as a clutch for selecting the flow stream, which stops once the participant's finger is released. Dragging also allows for 3 degree of freedom control since translation of the smartphone results in a corresponding translation of flow.

To move the airflow with *button*, the user presses a directional button on the screen to move the airflow at a fixed angular interval. For example, pressing the right-hand button would move that airflow to the right relative to the arrow pad.

## 3 EXPERIMENT

Here, we wanted to test which method could most quickly manipulate the air flows in the room to a desired target position. To do so, we set up an experiment with 50 participants. The general task for the experiment was to use our AR interface to move vent air flows towards a series of target cubes so that the resulting airflow output fell onto the cube.

The experiment was set up in a large office building with over 50 employees in one of the large office spaces. 50 participants were recruited for the experiment (18 female and 32 male, average age of 37.24, stdev. 11.55) as approved by an institutional review board (IRB). Participants used a touch-based Android smartphone to conduct the experiment.

### 3.1 Procedures

After first explaining the experiment task, the user engaged in 8-10 minutes of training with one of the three methods (the order was randomized between participants) to learn how to control and manipulate air flow with that particular method. Once the training was completed, participants engaged in the experiment and aligned air vents to virtual cubes. Upon completing all alignments for the 3rd method and hence the entire experiment, participants were given a survey with questions about the ease of use of the interfaces, rated on a 7-point Likert scale.

### 3.2 Results and Discussion

Our primary result from the experiment was that swipe outperformed drag and button on completion times, which was statistically significant. Swipe was also ranked highest as the preferred method but received the highest score for complexity, though this was not significant. All data was analyzed with analysis of variance (ANOVA) for quantitative data and with a Kruskal-Wallis test for subjective results. A summary of results is shown in Figure 2.

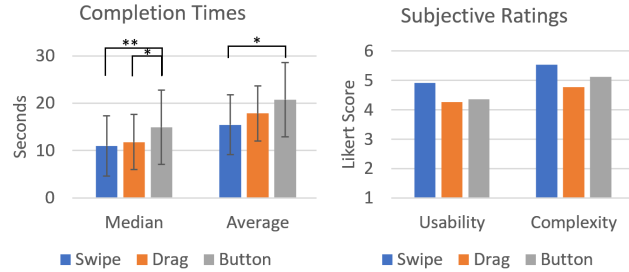


Figure 2: Results showing completion time per method (left) and subjective scores for usability and complexity (right) on a Likert scale from 1 to 7. \* and \*\* represent significance levels of 0.05 and 0.01.

We found a significant difference in median completion times ( $F = 3.058$ ,  $P < 0.01$ ) between the drag, swipe, and button methods, which were 11.81, 10.98, and 14.94 seconds, respectively. We also followed up with post-hoc t-tests, which showed that drag outperformed button ( $t = 2.27$ ,  $p < 0.05$ ) and that swipe outperformed button ( $t = 2.78$ ,  $p < 0.01$ ) for median completion times. No difference was found between drag and swipe ( $t = 0.25$ ,  $P = 0.50$ ). We did the same for average completion times, and swipe outperformed button ( $t = 2.89$ ,  $p < 0.01$ ), but differences for drag vs button and swipe were insignificant ( $t = 1.49$ ,  $P = 0.14$ ) and ( $t = 1.27$ ,  $p = 0.21$ ), respectively.

In addition to completion times, we looked at the results of the surveys we gave to participants after finishing the experiment. These contained two items rated via Likert scale, including 1) the ability to use as expected and 2) complexity. For the first item, users rated the drag, swipe, and button interfaces as 4.26, 4.91, 4.36, respectively. These results were not significant (Kruskal-Wallis:  $H=4.22$  and  $p = 0.12$ ). The results for the second question regarding interaction complexity were almost significant, with average ratings of 4.77, 5.53, and 5.12, for drag, swipe, and button, respectively (Kruskal-Wallis:  $H = 4.92$  and  $p = 0.085$ ).

## 4 CONCLUSION

In this paper, we implemented a system that allows for real time visualization and control of air conditioning air flow on a smartphone. In addition, we implemented three different methods for controlling the air flow trajectories, including drag, swipe, and button. In an experiment testing the performance of each of these methods, we found that the swipe method resulted in the quickest completion times, but that no method significantly outperformed any other on subjective scoring.

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